Problem-Solution Fit

Date	15 March 2026
Team ID	SWTID-2026-9848
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS]
• New joiners and employees needing a standard laptop
• Reporting managers who approve requests
• IT procurement / fulfilment team
• IT service desk and asset managers

6. CUSTOMER LIMITATIONS [CL]
EG. BUDGET, DEVICES
• Limited IT budget and fixed standard models
• Small IT team, limited time for manual work
• Needs to work on existing ServiceNow platform
• Varied staff comfort with new tools

5. AVAILABLE SOLUTIONS [AS]
PROS & CONS
• Email / phone requests: easy, but untracked and slow
• Spreadsheets: cheap, but error-prone, no approvals
• Manual catalog fulfilment: tracked, but many manual steps
• Custom scripting: flexible, but hard to maintain

2. PROBLEMS / PAINS + ITS FREQUENCY [PR]
• Manual laptop ordering delays delivery (every new hire / refresh)
• Approvals get stuck or forgotten (frequent)
• Requesters lack status visibility (every request)
• Repeated data entry mistakes (often)

9. PROBLEM ROOT / CAUSE [RC]
• No automated workflow for standard laptop requests
• Approval and task handoffs done by hand
• Requests spread across email and spreadsheets
• No single place for tracking and notifications

7. BEHAVIOR + ITS INTENSITY [BE]
• Employees chase IT by email/calls (high)
• Managers approve late or miss requests (medium)
• IT re-enters the same details in several places (high)
• Repeated follow-ups for status (high)

3. TRIGGERS TO ACT [TR]
• New employee onboarding
• Laptop refresh cycle or damaged device
• Long delays and rising complaints
• Management push for automation

10. YOUR SOLUTION [SL]
• A Service Catalog item "Standard Laptop Request" on ServiceNow, with the process automated by Flow Designer.
• On submission, the flow validates the request and sends it automatically to the manager for approval.
• After approval, it creates catalog tasks for procurement and delivery, and updates request status.
• Automatic email notifications keep requester, approver and IT team informed at every step.
• All requests, tasks and assets are stored in platform tables, giving full tracking and audit.
• Result: faster orders, fewer errors, less manual effort and complete visibility.

8. CHANNELS OF BEHAVIOR [CH]
ONLINE
• Service Portal / Employee Center request form
• Email notifications and approvals
• Now Mobile app

4. EMOTIONS BEFORE / AFTER [EM]
• Before: frustrated, anxious, uncertain
• After: relieved, confident, in control

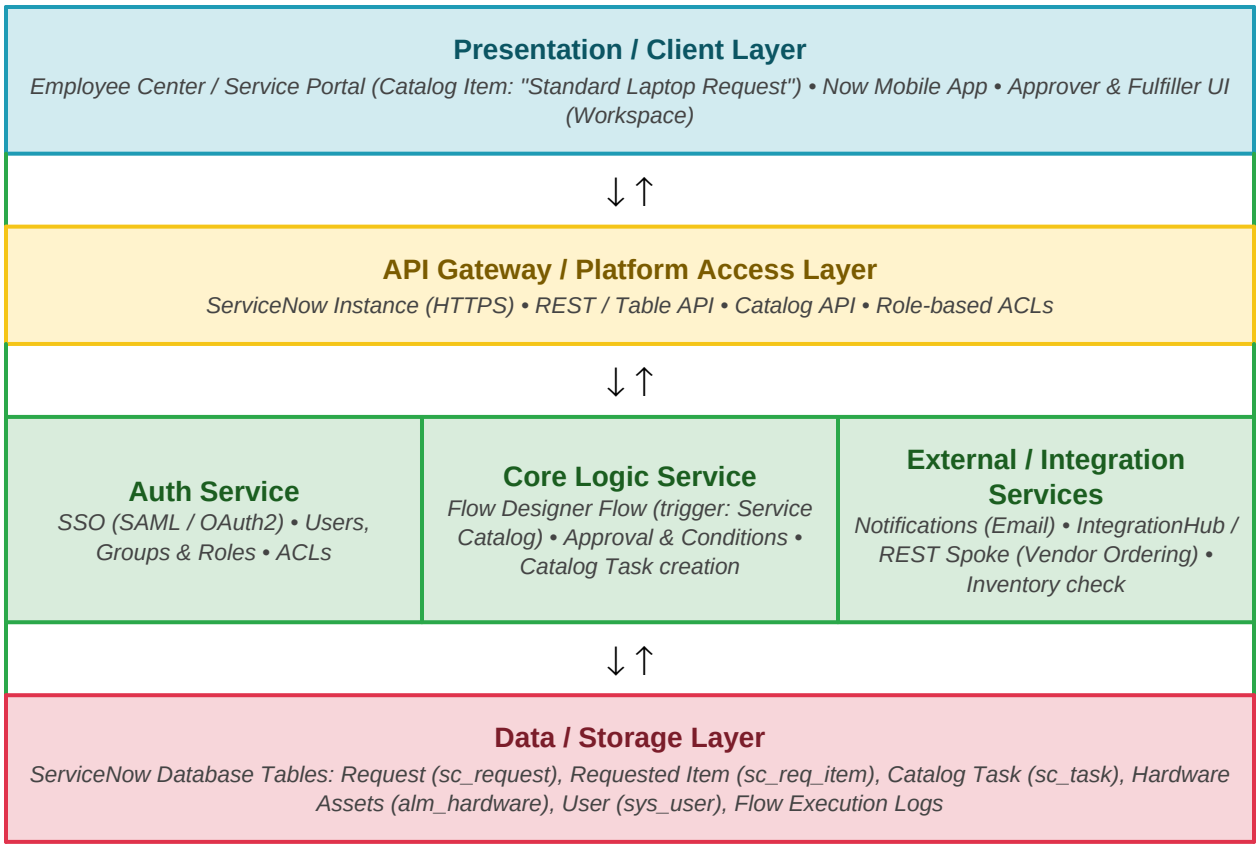
OFFLINE
• Verbal follow-up with manager / IT desk
• Physical laptop handover and sign-off

Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-9848
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture of the automated standard laptop ordering solution built on the ServiceNow platform using Flow Designer, organised into Presentation, Application and Data layers.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Employees browse the catalog and submit a Standard Laptop Request through a simple form. Managers approve and IT fulfillers track tasks from their own views.	ServiceNow Employee Center / Service Portal, Service Catalog, Record Producer / Catalog Item, Now Mobile
API Gateway	Single secure entry point to the instance. Handles HTTPS requests, authenticates callers and enforces role-based access to catalog and table data.	ServiceNow REST Table API, Catalog API, ACLs, HTTPS
Auth Service	Verifies user identity and permissions so only authorized employees can order and only approvers can approve.	SSO (SAML / OAuth2), Users, Groups, Roles, ACLs
Microservices (Core Logic)	Flow Designer flow triggers when a laptop request is submitted: validates the request, sends it for manager approval, and on approval creates catalog tasks for procurement and delivery, updating request status automatically.	Flow Designer, Flow Actions, Approval Rules, Catalog Tasks, Flow Variables
External Integrations	Sends email notifications to requester, approver and IT team, and can place the order with the vendor / check inventory through outbound calls.	Notification Engine (Email), IntegrationHub, REST Spoke, Vendor API
Database	Stores requests, requested items, tasks, users, assets and flow execution history for tracking, reporting and audit.	ServiceNow Platform Database, Tables: sc_request, sc_req_item, sc_task, alm_hardware, sys_user

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	SWTID-2026-9848
Project Title	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform standard laptop procurement by automating the request, approval and fulfilment process with ServiceNow Flow Designer. It tackles manual delays and untracked requests, promising faster delivery, fewer errors and better visibility for employees, managers and the IT team. Key features include an automated approval flow and real-time status notifications.

Project Overview	
Objective	The primary objective is to automate standard laptop ordering using Flow Designer, so that requests are approved and fulfilled faster with minimal manual effort.
Scope	The project covers the end-to-end standard laptop request process on ServiceNow: catalog request submission, manager approval, procurement and delivery tasks, notifications and status tracking.
Problem Statement	
Description	Standard laptop orders are handled through manual emails and follow-ups. This causes delayed approvals, repeated data entry, poor status visibility and slow delivery, affecting IT efficiency and employee experience.
Impact	Solving these issues will result in faster laptop delivery, fewer errors, reduced manual workload for IT, and an overall improvement in the IT procurement process and employee satisfaction.
Proposed Solution	
Approach	Building a Service Catalog item for standard laptop requests and using Flow Designer to automate approvals, task creation and notifications on the ServiceNow platform.
Key Features	- Service Catalog item for standard laptop requests. - Automated manager approval flow with conditions. - Automatic creation of procurement and delivery tasks. - Real-time email notifications and request status tracking.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard laptop, Intel i5 (4 cores), no GPU needed
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD (data hosted on ServiceNow cloud instance)
Software		
Platform / Frameworks	Low-code platform	ServiceNow Personal Developer Instance (PDI), Flow Designer, Service Catalog
Libraries	Additional libraries	Built-in: Approval Engine, Notification Engine, Flow Actions, IntegrationHub
Development Environment	IDE	Web browser (Google Chrome) with ServiceNow Studio
Data		
Data	Source, size, format	Test data created in ServiceNow instance, about 10-20 sample requests, platform tables